

BENCHAO SU

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ACADEMIC INTERESTS

My research interest lies in the p -adic local Langlands program, especially in its geometric aspects.

EDUCATION

Peking University

- Ph.D. in Pure Mathematics (expected graduation: 2026).
- Advisor: Yiwen Ding.

PUBLICATIONS AND PREPRINTS

3. **De Rham cohomology of Lubin-Tate spaces and Drinfeld spaces.** arXiv:2503.09352.
2. **On the locally analytic Ext^1 -conjecture in the $\mathrm{GL}_2(L)$ case.** arXiv:2504.17683.
1. **Locally analytic vectors in the completed cohomology of unitary Shimura curves** (joint with Qiu Tian). arXiv:2505.10290.

TALKS

5. *Solid locally analytic representations and solid almost quasi-coherent sheaves*, Recent progress in p -adic geometry, 2025.03, Yunnan.
4. *On Breuil's locally analytic Ext^1 -conjecture in the $\mathrm{GL}_2(L)$ case*, ENTR Workshop 24, 2024.09, Essen.
3. *Translations and the locally analytic Ext^1 conjecture in the $\mathrm{GL}_2(L)$ case*, p -adic geometry and analysis, 2024.06, Shenzhen.
2. *Translations and the locally analytic Ext^1 conjecture in the $\mathrm{GL}_2(L)$ case*, p -adic Day, 2024.05, Beijing.
1. *Locally σ -analytic vectors in the completed cohomology of unitary Shimura curves*, Mini-conference on p -adic Langlands Program, 2023.08, Beijing.

¹Updated October 13, 2025